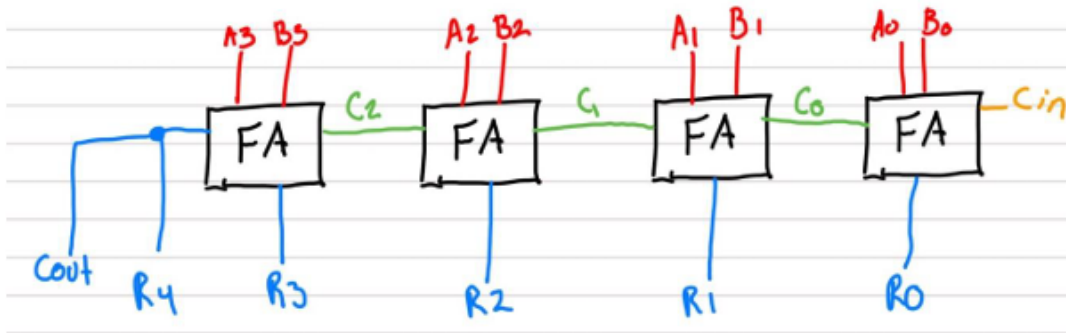


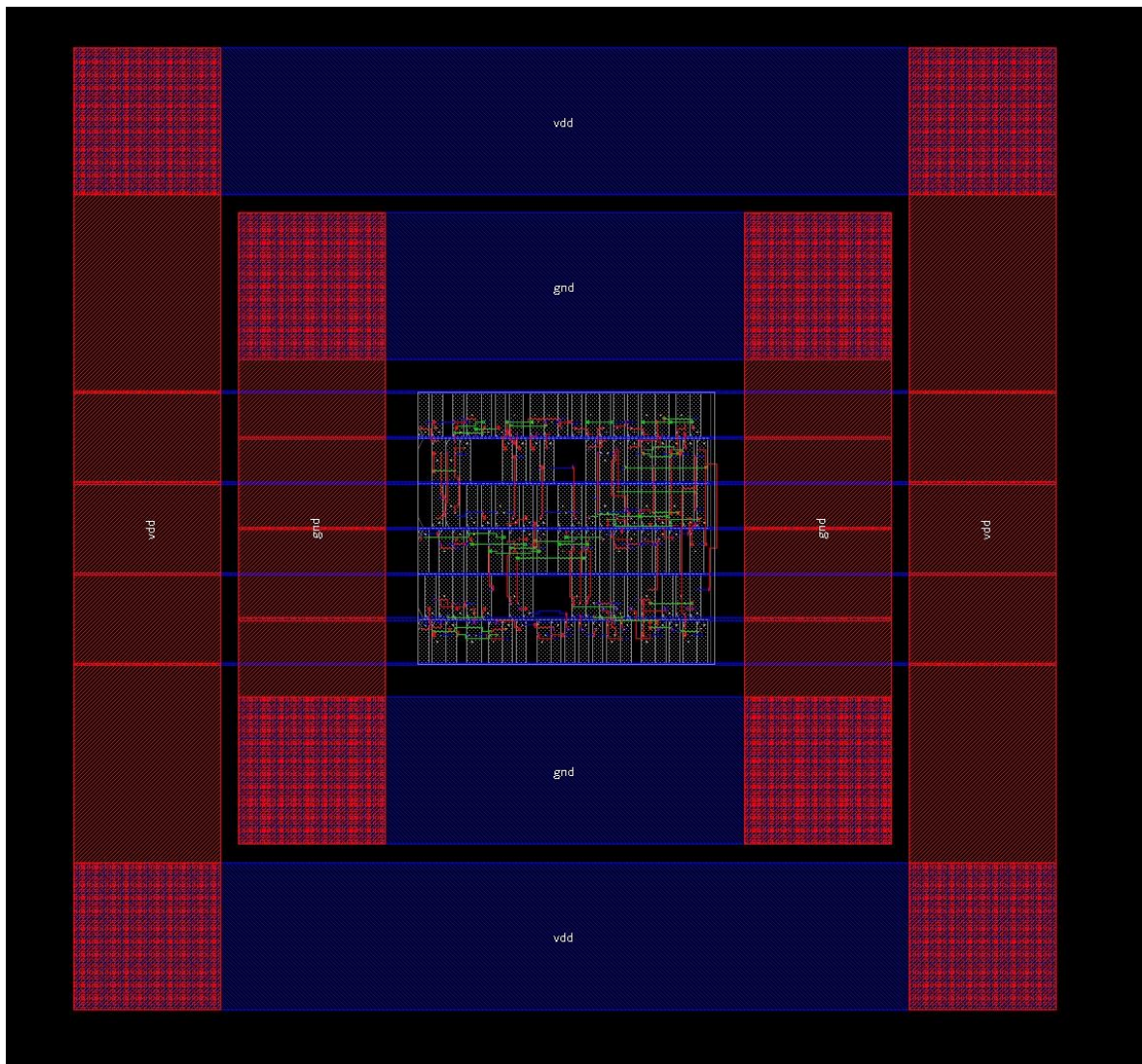
Practice #9: Generating layouts with Innovus

From the next description we create the layout using Innovus.

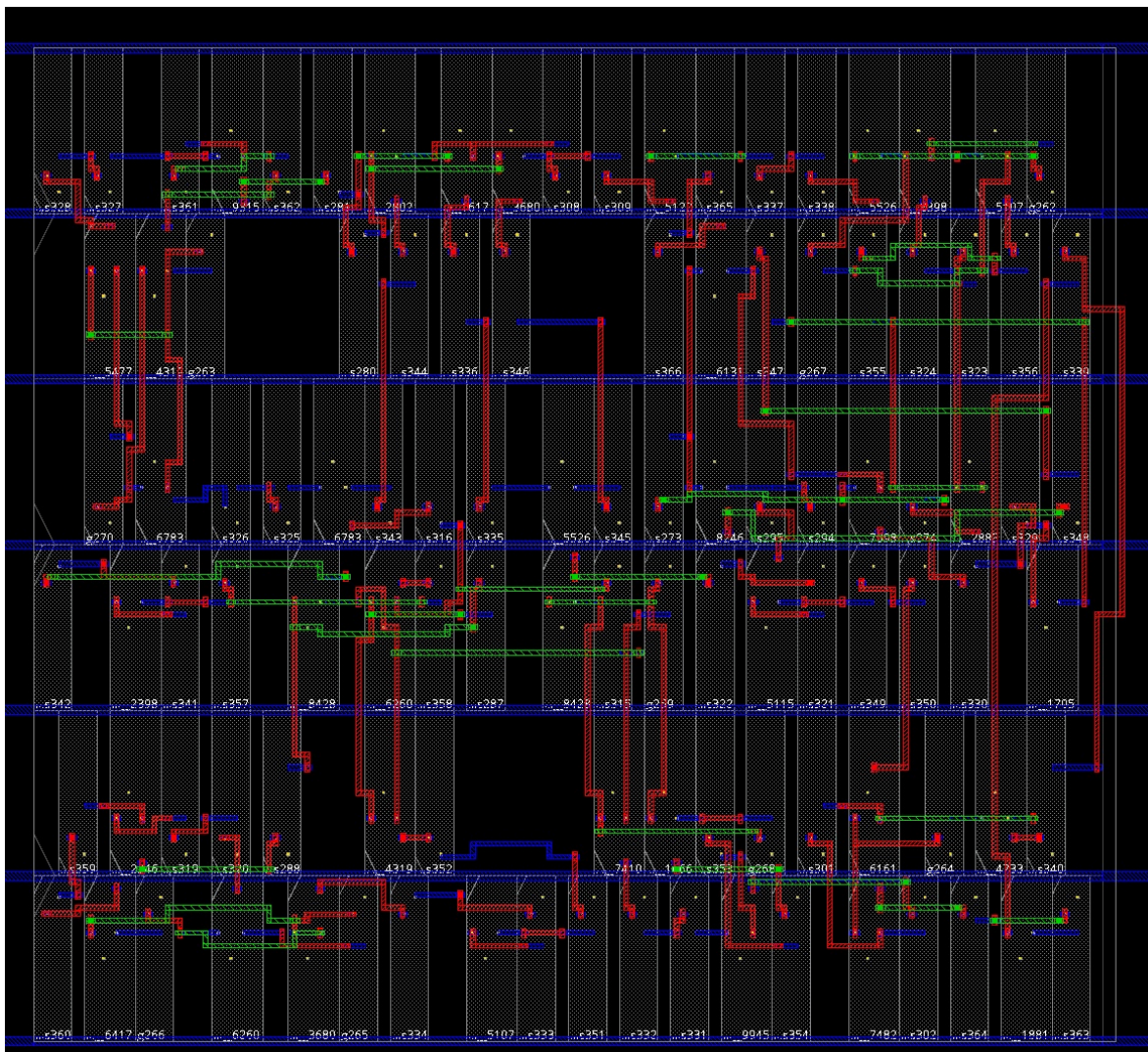


```
1 //Verilog HDL for "Verilog", "Full_Adder_4b" "functional"
2
3 module Full_Adder_4b (AM,BM,CinM,Result,CoutM);
4
5 input [3:0] AM,BM;
6 input CinM;
7 output [4:0] Result;
8 output CoutM;
9 wire [2:0] CoutS;
10
11 FA U0(.A(AM[0]),.B(BM[0]),.Cin(CinM),.S(Result[0]),.Cout(CoutS[0]));
12 FA U1(.A(AM[1]),.B(BM[1]),.Cin(CoutS[0]),.S(Result[1]),.Cout(CoutS[1]));
13 FA U2(.A(AM[2]),.B(BM[2]),.Cin(CoutS[1]),.S(Result[2]),.Cout(CoutS[2]));
14 FA U3(.A(AM[3]),.B(BM[3]),.Cin(CoutS[2]),.S(Result[3]),.Cout(Result[4]));
15
16 assign CoutM = Result[4];
17
18 endmodule
19
20 //Full Adder Module
21 module FA (A,B,Cin,S,Cout);
22
23 input A,B,Cin;
24 output S,Cout;
25
26 assign S = ((A ^ B) ^ Cin);
27 assign Cout = (((A ^ B) & Cin) | (A & B));
28
29 endmodule
30
```

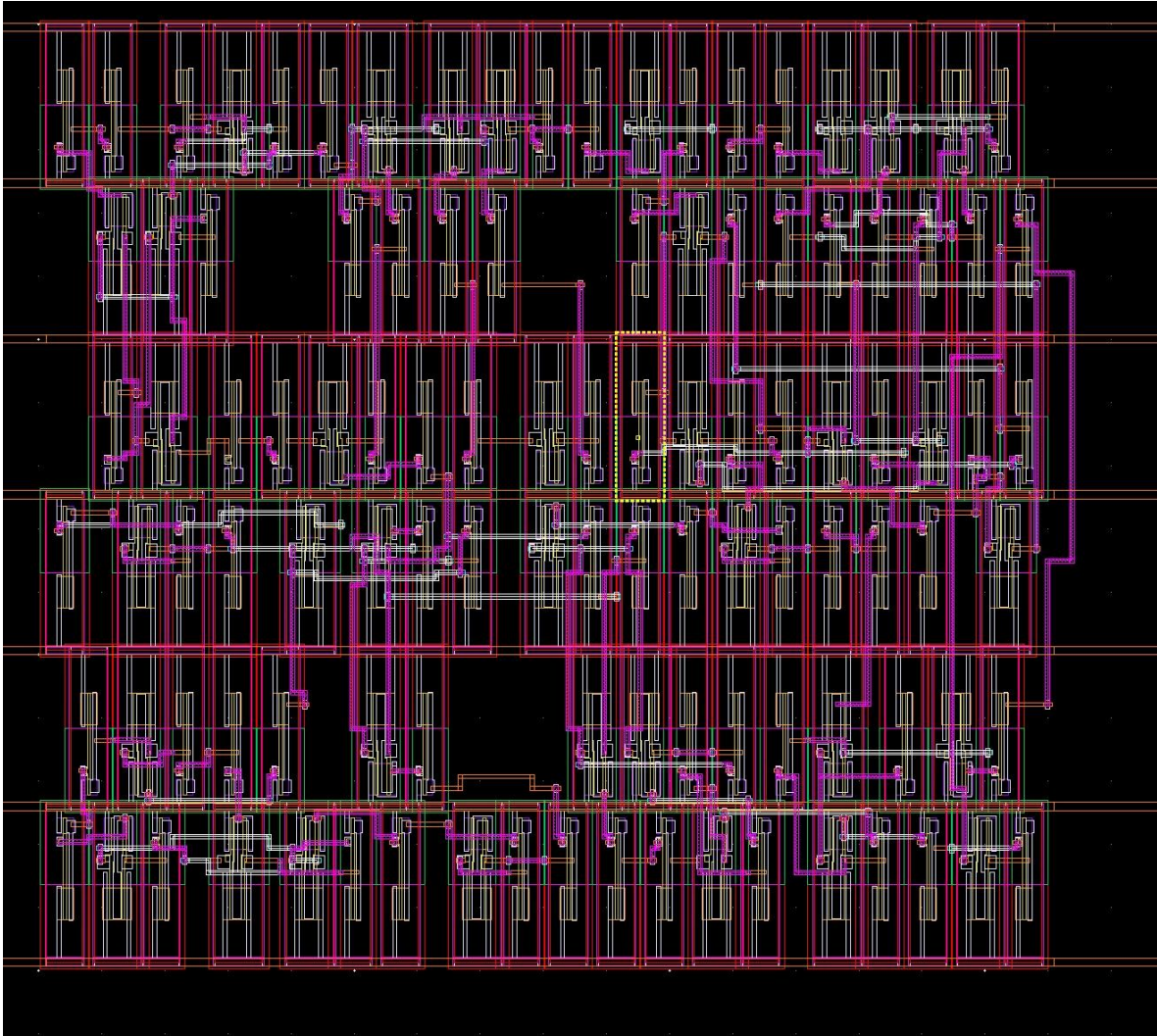
When we use the Innovus tool we get the next layout.



If we zoom in to see the actual circuit, we can see it better.



But that is only using Innovus if we export it to Virtuoso, we have the following layout.



As we can see if we see carefully the layouts we can see that essentially it's the same layout and it used the same basic cells we provide in the library, we can the names from each component but I decide to remove them cause I have a lot of standard cells and all the text of the name doesn't let us to see how the layout looks like.

